

I specialise in architecting production-grade, agentic AI systems on secure Azure PaaS foundations that help organisations adopt AI safely and scale reliably, backed by Microsoft certifications as an Azure AI Engineer Associate and Azure Solutions Architect Expert. In my recent role, I addressed the challenge of moving generative AI initiatives beyond proof-of-concept, the organisation had scattered LLM experiments, no consistent RAG or agent architecture, and no governed PaaS foundation to build them on.

I led the design of a multi-agent AI platform built on Microsoft Foundry, Microsoft Agent Framework, Semantic Kernel, LangChain, LangGraph, LangSmith, AutoGen and CrewAI, integrated with MCP (Model Context Protocol) for standardised tool and context access across agents. After assessing use cases across document processing, search and workflow automation, I defined reusable patterns for RAG pipelines, vector search, agent memory and function-calling tools, so teams could compose new agentic workflows without rebuilding core infrastructure each time.

Underpinning this, I designed the Azure PaaS architecture the AI platform runs on App Service, Azure Functions, API Management, Logic Apps, Azure Kubernetes Service, Container Apps, Event Grid, Event Hub, Service Bus, Azure SQL, Cosmos DB, Storage and Key Vault — selecting each service and integration pattern based on resilience, latency, coupling, scalability and recovery requirements. I applied landing zone principles, private connectivity and centrally managed security controls so that both traditional workloads and AI services shared a single, governed, well-architected foundation rather than parallel, inconsistent environments.

I made safety and evaluation a design prerequisite rather than an afterthought. I embedded Azure AI Content Safety, guardrails and structured evaluation frameworks directly into the agent pipeline, alongside model routing to balance cost, latency and capability across multiple LLMs, and drew on Hugging Face models, fine-tuning and prompt engineering to adapt capabilities to specific use cases. I also established reference architectures spanning Azure AI Search, Document Intelligence, Vision, Speech, Language and Translator services, with key architectural decisions documented as Architecture Decision Records.

I worked closely with engineering, security, data and business stakeholders to align on architecture, resolve integration and compliance concerns, and support rollout. By embedding safety and governance into reusable PaaS and agent components, teams could ship new AI capabilities without re-litigating security or compliance for every project.

The platform reduced time-to-production for new AI use cases, improved deployment consistency across PaaS environments, cut LLM inference costs through model

routing, and gave the organisation a governed, reusable foundation for scaling both AI and digital services enterprise-wide.